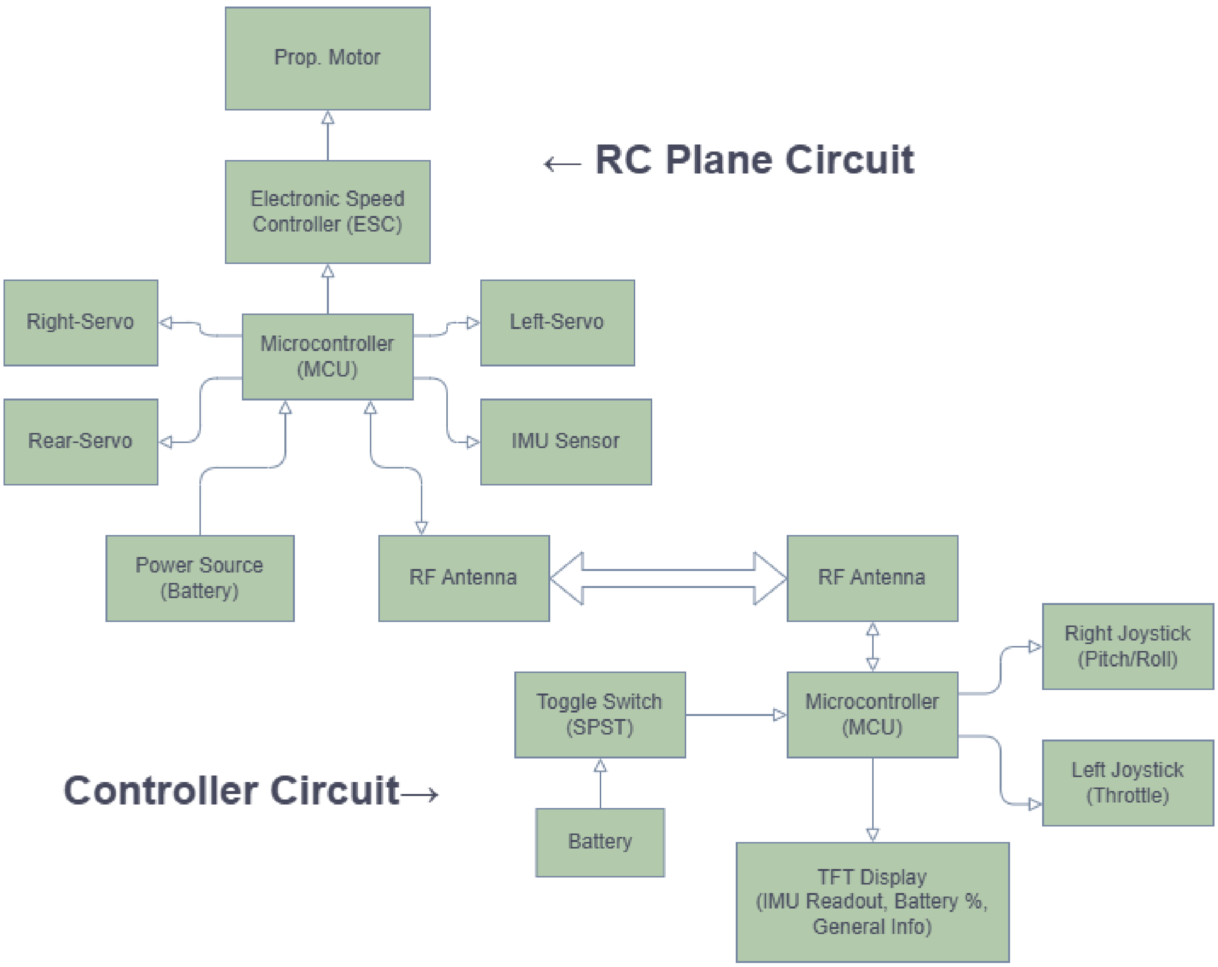
ENGR 2999 Progress Report 2

Project Recap

Recall, for our ENGR2999 Capstone project we will be designing and fabricating a RC airplane that responds to the user interface of a joystick and throttle to control motor power and flap control. Our project block diagram is shown below.



Progress Updates

Fuselage

We have completed the design of the fuselage of the airframe. This has taken longer than expected as we have had to change the original plan of 3d printing the fuselage which failed and have now pivoted to a laser cut plywood approach shown both below. One upside of the finger jointed plywood over the 3d print is it provides superior torsional rigidity for our powerful motor.

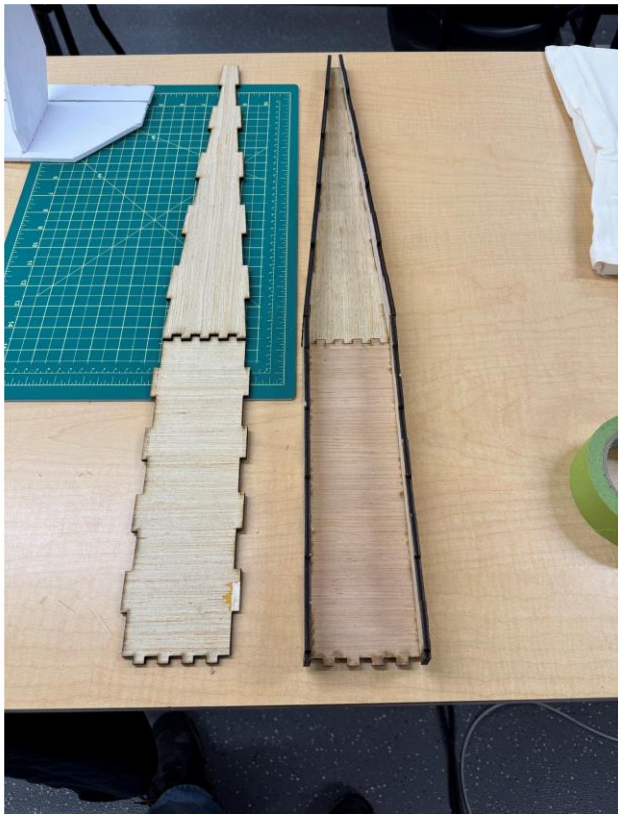


Figure 1: Laser cut plywood fuselage

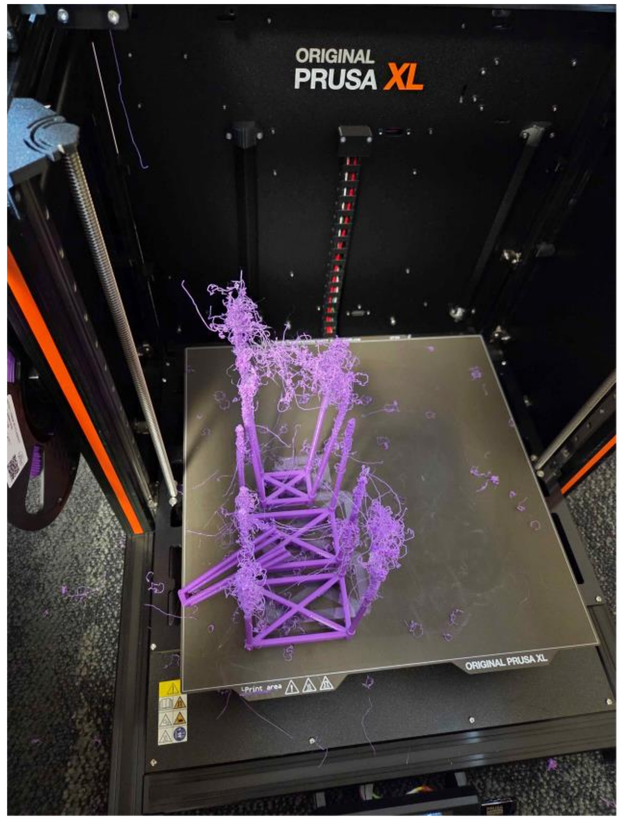


Figure 2: Failed 3d print of fuselage

Tail Assembly

We have completed initial the design for the horizontal and vertical stabilizers which connect to a servo motor within the fuselage for control of the elevator to control altitude during flight. Shown below is how the system will sit in the plane and how it all connects.

Figure 3: Servo and stabilizer (top view)

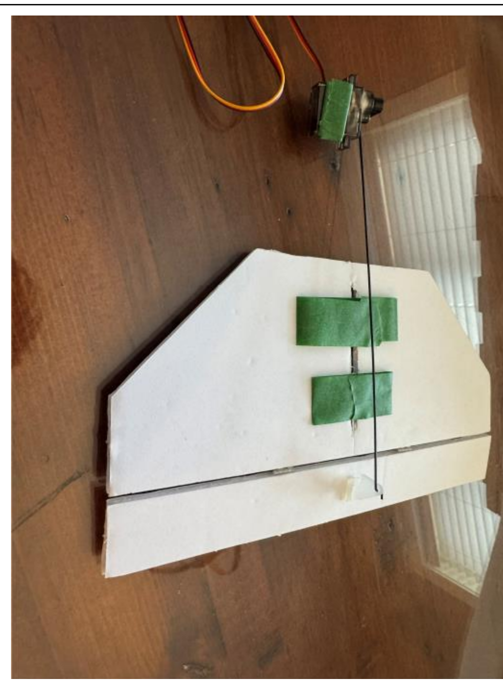


Figure 4: Servo and stabilizer (bottom view)

Power Train

Our aircraft will be powered by a DYS D3536 brushless motor connected to a 60A DYS ESC and a Melasta 4S 14.8 V battery. This power unit combo was selected to give us plenty of wiggle room in the weight of the aircraft and now that the heavier plywood is being used perfectly demonstrates why we went with the more powerful setup. The green masking tape seen in the photos is for initial prototyping and layout, and that final assembly will use mechanical fasteners or epoxy to ensure high-vibration resistance.

	
Figure 5: Motor, battery, and ESC (prototype)	**Figure 6:** Motor mount

Control Board

The control board is based around the Waveshare RP2350-Zero microcontroller which interfaces with the NRF24L01 antenna via SPI and the MPU6050 gyroscope + accelerometer via I2C. The control board is soldered and hand wired to reduce its footprint and weight. The board features 4 servo control channels with the main motor channel providing the board with a 5 volt source through the ESC's Battery Eliminator Circuit (BEC). The BEC connector follows the same pinout as a servo connector with the only difference being that it's a power source instead of requiring power. The antenna has a detached breakout board allowing it to be mounted anywhere within the plane at the cost of higher signal interference, which will be resolved with cable shielding. The board is complete and tested but the new firmware still needs to be written.

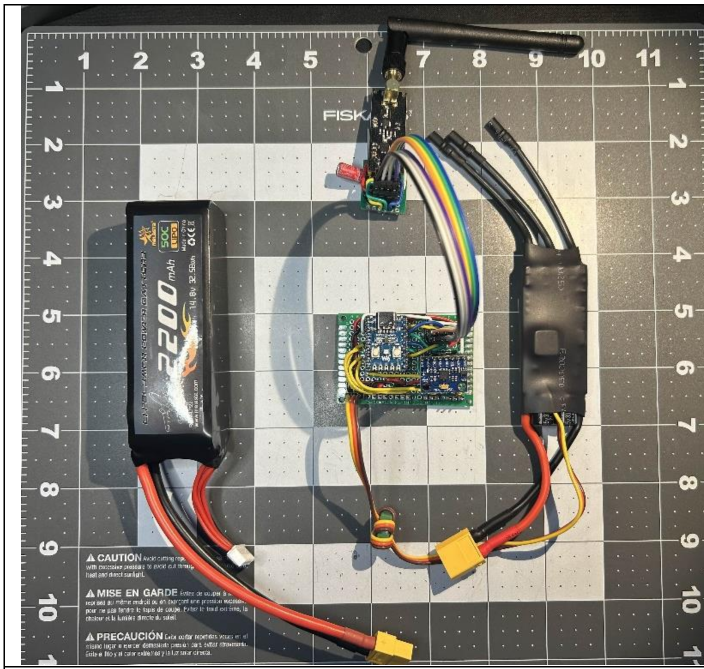
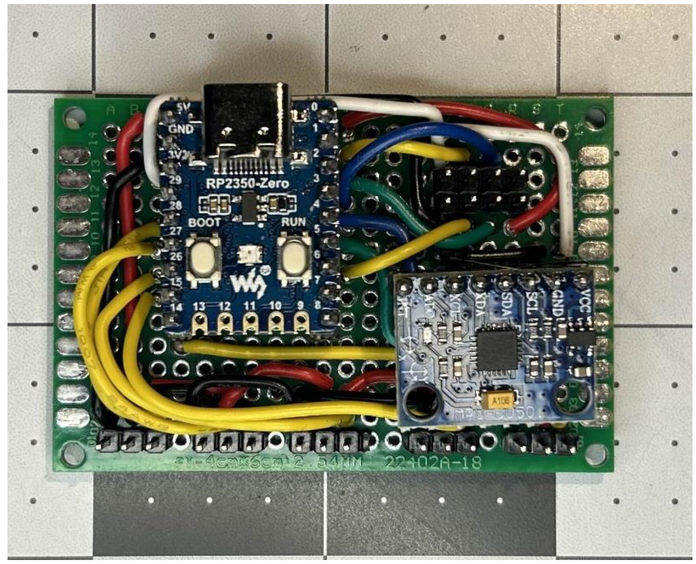
	

Figure 7: Mock Circuit With Control Board	**Figure 8:** Control Board Closeup

Controller

This is the second iteration of the controller, which is running the same firmware as the first. The controller is based around the Waveshare RP2350-Zero, using a matching NRF24L01 antenna and an SSD1306 OLED display. The control surface is the same as the first, with the addition of a rotary encoder. The controls are as follows: the left slide potentiometer is for throttle control, the joystick on the right is for pitch and roll control, and the rotary encoder is for internal controls such as trim and servo range. The rear of the controller has all the power circuitry, which is as follows: a single salvaged LiPo battery is connected to a TP4056 battery charging module, with a switch between the output of the battery/charger and an XL6009E1 boost converter, which feeds the board 5V. The 5V line then goes through the MCU's dropout regulator to supply all the 3V3 components on the board. This is the final version of the controller and is surprisingly comfortable to hold.

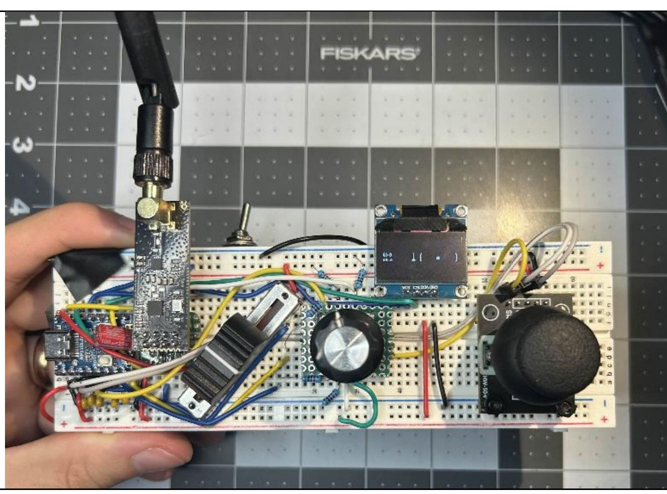
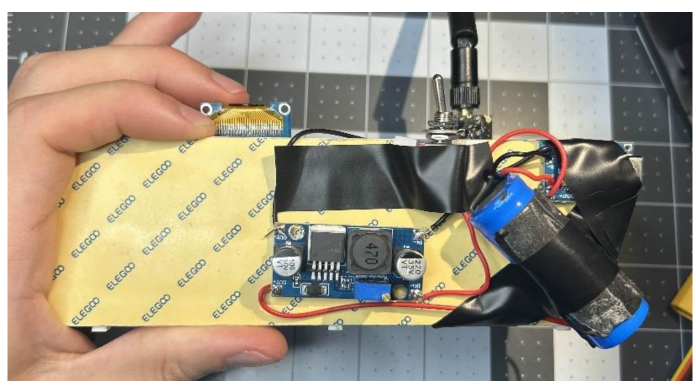
	

Figure 9: Final Controller Design (front)	**Figure 9:** Final Controller Design (back)

Software

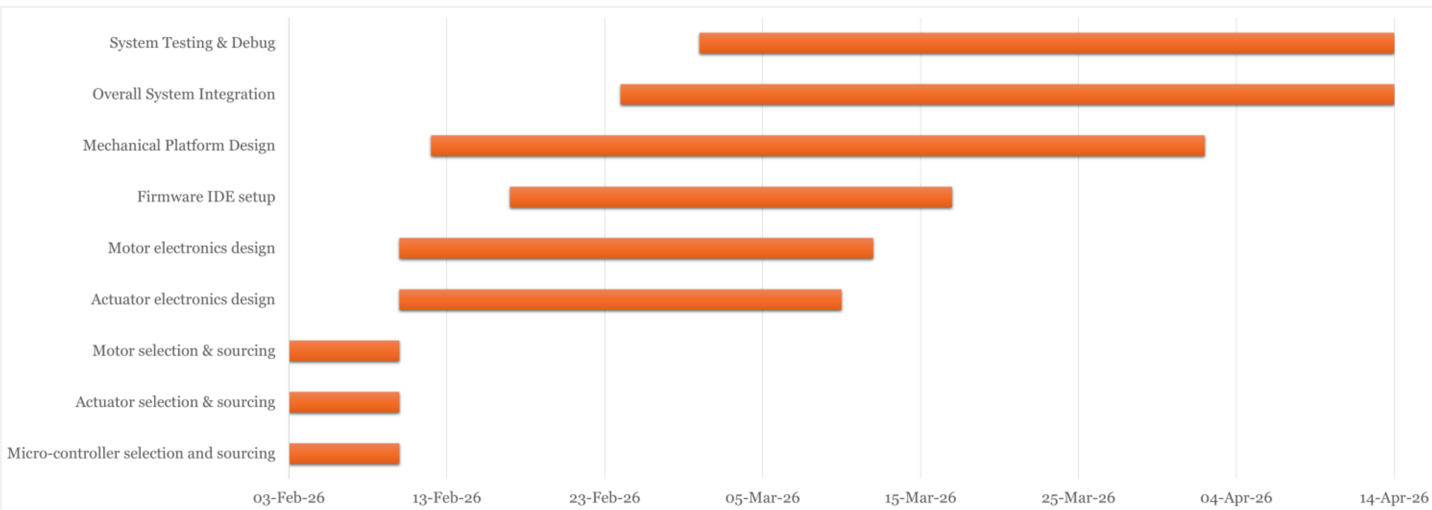
The firmware has gone through several iterations, with the current version rolled back to the stable first version for testing. The next and final version of the firmware will be a slimmed-down version for the sake of time and simplicity. Currently, the display drawing, data and input handling, and trim/range control all need to be rewritten to match the final expected features. Overall, the software is very near completion, and with the final control board and controller completed, the firmware will be very straightforward to write.

Schedule Updates

So far, our schedule has had the following changes:

1. The 3d printing mishap has moved our Mechanical Platform Design deadline to 02 – Apr – 2026
2. Because the mechanical design is taking longer than expected, system integration and system testing has been pushed to 04 – Apr – 2026

An updated Gantt chart with the above changes is shown below



Problem Summary

1. Mechanical Design is taking much longer than anticipated. The initial plan was to 3d print most of the aircraft. However, with the deadline fast approaching coming up with a new approach after our failed 3d print was needed and have now switched to a laser cut plywood approach to meet the deadline by 02 – Apr – 2026.
2. As of 02 – Apr – 2026, the calculated take-off weight is approximately 1100g. With our 1450KV motor producing approximately 1500g of thrust, we have maintained a 1.36:1 thrust-to-weight ratio, ensuring sufficient power for a safe climb-out during the maiden flight.
3. Now our project is on track to reach a working prototype by 14 – Apr – 2026. However, a known unknown we have is how much time we will get to test as we might have a scenario where we reach 14 – Apr – 2026. To mitigate the risk of a single-point-of-failure during the April 14th flight, we are prioritizing 'dry-run' ground testing and electronic bench-testing to ensure system reliability before the first takeoff.]
4. While further testing is required, the actual range of the antenna has appeared to be significantly less than advertised, potentially related to the intended use case of the antenna. It was found that the connection stability is significantly related to the relative orientation of the antennae, with alignment being much more important than anticipated, this will be investigated further.
5. A propeller solution has still not been found and will need to be resolved before the end of assembly.